# Linux Basics

# Day- 1

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## What is Linux?

- Linux is an operating system like windows, macos.
- It’s open-source means anyone can use, modify and share it for free.
- It is used widely for its stability, security and flexibility.

## Linux Distribution

- It is a version of linux that combines the linux kernel with additional software, tools and a user interface to create a complete operating system.

Example:-

- Ubuntu
- Fedora
- Kali linux
- Rhel
- CentOS

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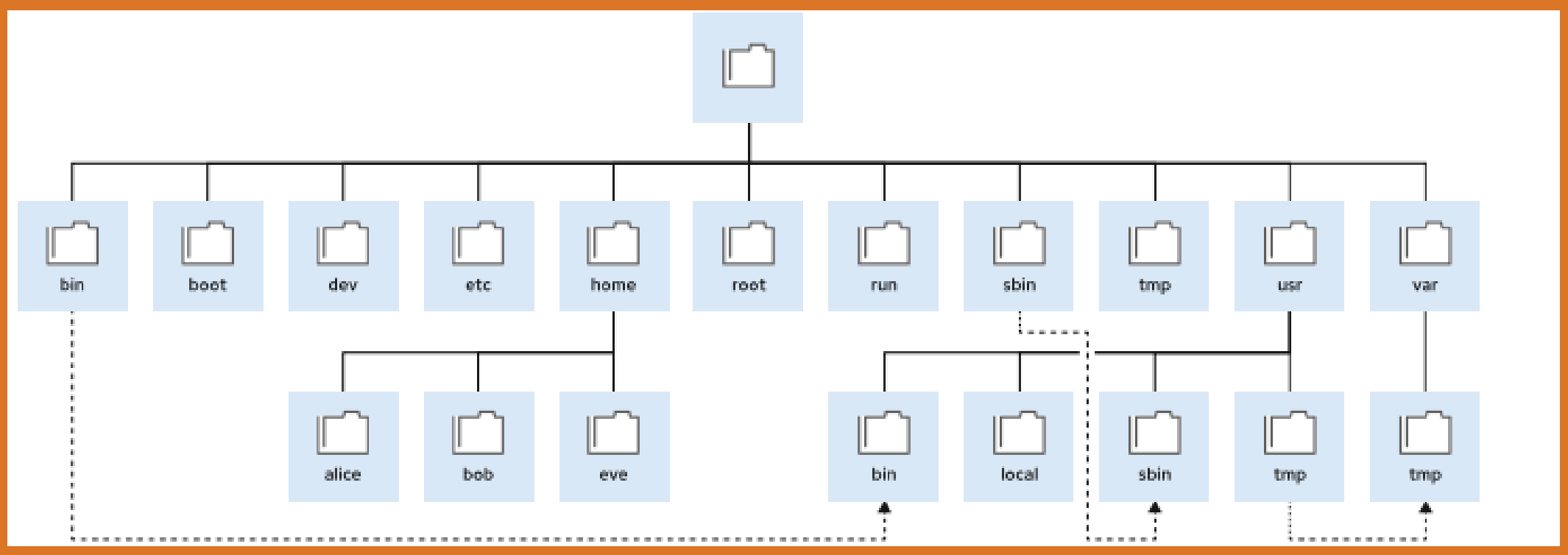
# Linux Architecture




- The Linux system stores all files on file systems, which are organized into a single inverted tree known as a file-system hierarchy.
- The / directory is the root directory at the top of the file-system hierarchy.

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| Location | Purpose |
| --- | --- |
| /boot: | Files to start the boot process. |
| /dev: | Special device files that the system uses to access hardware. |
| /etc: | System-specific configuration files. |
| /home: | Home directory, where regular users store their data and configuration files. |
| /root: | Home directory for the administrative superuser, root. |
| /run: | Runtime data for processes that started since the last boot. This data includes process ID files and lock files. |

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/tmp: Files that are not accessed, changed, or modified for 10 days are deleted from this directory automatically.

/usr: Installed software, shared libraries, including files, and read-only program data.
Significant subdirectories include:
- /usr/bin: User commands
- /usr/sbin: System administration commands
- /usr/local: Locally customized software.

/var: used for files that grow and change over time, such as logs, caches, and temporary files.

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